1020 U2 mutually exclusive and indep.

1. A high school science teacher has 78 students. Of those students, 35 are in the band and 32 are on a sports team. There are 16 students who are not in the band or on a sports team. One student from the 78 students will be selected at random. Let event $\overline{B}$ represent the event of selecting a student in the band, and let event S represent the event of selecting a student on a sports team.

Are B and S mutually exclusive events?

(A) No, because $P(B \cap S) = \frac{5}{78}$. ✓

(B) No, because $P(B \cap S) = \frac{48}{78}$.

(C) No, because $P(B \cap S) = \frac{62}{78}$.

(D) Yes, because $P(B \cap S) = \frac{5}{78}$.

Answer A

Correct. The sum of 35, 32, and 16 is 83, which is 5 greater than 78. So there must be 5 students who are both in the band and on a sports team. Because the events can occur at the same time, they are not mutually exclusive.

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2.

	Team		
	Home	**Away**	**Total**
Purchased food	120	40	160
Did not purchase food	60	30	90
Total	180	70	250

The table shows data that were collected from people who attended a certain high school basketball game and indicates the team each person rooted for and whether each of these people purchased food during the game. A person who attended the game will be selected at random. Which of the following correctly interprets mutually exclusive events represented by the table?

(A) Rooting for the home team and rooting for the away team ✓

(B) Rooting for the home team and purchasing food during the game

(C) Rooting for the home team and not purchasing food during the game

(D) Not rooting for the home team and not purchasing food during the game

Answer A

Correct. Based on the table, each person either rooted for the home team or rooted for the away team. Because the events cannot occur at the same time, the events are mutually exclusive.

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3. The following table shows the classification of all 51 dogs from an animal shelter by whether the dogs lived mostly outdoors or mostly indoors before coming to the shelter and whether they tested positive or negative for a certain skin condition.

	Positive	**Negative**	**Total**
Mostly Outdoors	10	20	30
Mostly Indoors	8	13	21
Total	18	33	51

One dog from the animal shelter will be selected at random. Based on the table, which of the following statements is correct?

(A) The probability of selecting a dog that lived mostly indoors is greater than 0.5.

(B) The probability of selecting a dog that tested positive is greater than the probability of selecting a dog that tested negative.

(C) The probability of selecting a dog that tested positive given that the dog lived mostly outdoors is less than the probability of selecting a dog that tested positive given that the dog lived mostly indoors. ✓

(D) The probability of selecting a dog that lived mostly outdoors given that the dog tested positive is greater than the probability of selecting a dog that lived mostly outdoors given that the dog tested negative.

Answer C

Correct. The probability of selecting a dog that tested positive given that the dog lived mostly outdoors is $\frac{10}{30} = 0.333$, which is less than the probability of selecting a dog that tested positive given that the dog lived mostly indoors, $\frac{8}{21} = 0.381$.

4. Events D and E are independent, with $P(D) = 0.6$ and $P(D \text{ and } E) = 0.18$. Which of the following is true?

(A) $P(E) = 0.12$

(B) $P(E) = 0.4$

(C) $P(D \text{ or } E) = 0.72$ ✓

(D) $P(D \text{ or } E) = 0.9$

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5. One student from a high school will be selected at random. Let A be the event that the selected student is a student athlete, and let B be the event that the selected student drives to school. If $P(A \cap B) = 0.08$ and $P(B|A) = 0.25$, what is the probability that the selected student will be a student athlete?

(A) 0.02

(B) 0.17

(C) 0.32 ✓

(D) 0.33

Answer C

Correct. By the multiplication rule, $P(A \cap B) = P(A) \cdot P(B|A)$, so $0.08 = P(A) \cdot 0.25$. Solving for $P(A)$ gives 0.32.

6. Each of the faces of a fair six-sided number cube is numbered with one of the numbers 1 through 6, with a different number appearing on each face. Two such number cubes will be tossed, and the sum of the numbers appearing on the faces that land up will be recorded. What is the probability that the sum will be 4, given that the sum is less than or equal to 6 ?

(A) 2/36

(B) 3/36

(C) 3/15 ✓

(D) 2/9

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7. As a promotion, the first 50 customers who entered a certain store at a mall were asked to choose from one of two discounts. The first discount choice was 20% off all purchases made that day. The second discount choice was 10% off all purchases for the week. Of those who received the discounts, 28 chose the first discount and 22 chose the second discount. One customer will be selected at random from those who received a discount. Let F represent the event that the selected person chose the first discount, and let S represent the event that the selected person chose the second discount.

Are F and S mutually exclusive events?

(A) Yes, because $P(F \cap S) = 0$. ✓

(B) Yes, because $P(F \cap S) = 0.12$.

(C) Yes, because $P(F \cap S) = 1$.

(D) No, because $P(F \cap S) = 0$.

Answer A

Correct. A person could choose only one discount. Because the events cannot occur at the same time, the events are mutually exclusive.

8. For flights from a particular airport in January, there is a 30 percent chance of a flight being delayed because of icy weather. If a flight is delayed because of icy weather, there is a 10 percent chance the flight will also be delayed because of a mechanical problem. If a flight is not delayed because of icy weather, there is a 5 percent chance that it will be delayed because of a mechanical problem. If one flight is selected at random from the airport in January, what is the probability that the flight selected will have at least one of the two types of delays?

(A) 0.065

(B) 0.335 ✓

(C) 0.350

(D) 0.450

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9. The probability that a particular electrical component operates successfully is $\frac{3}{4}$. An electrical system consists of three such components that operate independently. The system will function successfully if at least one of the three components operates successfully.

What is the probability that the system will function successfully?

(A) $\frac{1}{64}$

(B) $\frac{27}{64}$

(C) $\frac{37}{64}$

(D) $\frac{63}{64}$ ✓

Answer D

Correct. If F represents the probability that a component fails, then $\frac{63}{64}$ is the probability that at least one of the components operates successfully, or 1 minus the probability that all three components fail.
$1 - P(F_1 \cap F_2 \cap F_3) = 1 - P(F_1)*P(F_2)*P(F_3) = 1 - \left(\frac{1}{4}*\frac{1}{4}*\frac{1}{4}\right) = 1 - \frac{1}{64} = \frac{63}{64}$

10. Ali surveyed 200 students at a school and recorded the eye color and the gender of each student. Of the 80 male students who were surveyed, 60 had brown eyes. If eye color and gender are independent, how many female students surveyed would be expected to have brown eyes?

(A) 20

(B) 30

(C) 90 ✓

(D) 100

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11. A fair coin will be tossed three times. The set of possible outcomes may be described by the ordered triples
{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT}, where H represents heads and T represents tails.

What is the probability that all three tosses result in heads, given that the three tosses result in more heads than tails?

(A) $\frac{1}{8}$

(B) $\frac{1}{4}$ ✓

(C) $\frac{3}{8}$

(D) $\frac{1}{2}$

Answer B

Correct. The probability that all three tosses result in heads is $\frac{1}{8}$. The probability that all three tosses result in more heads than tails, the outcomes HHH, HHT, HTH, THH, is $\frac{4}{8}$. The conditional probability that all three tosses result in heads, given that the three tosses result in more heads than tails is $\frac{\frac{1}{8}}{\frac{4}{8}} = \frac{1}{4}$.

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12. The seniors at three high schools were surveyed about their plans after graduation. The following table shows the responses, classified by high school.

	Work	Military	College	Undecided	Total
High School A	99	49	138	63	349
High School B	62	26	156	54	298
High School C	83	31	124	71	309
Total	244	106	418	188	956

One senior from the high schools will be selected at random. What is the probability that the senior selected will not be from High School B given that the senior responded with a choice other than college?

(A) $\frac{538}{956}$

(B) $\frac{262}{418}$

(C) $\frac{658}{956}$

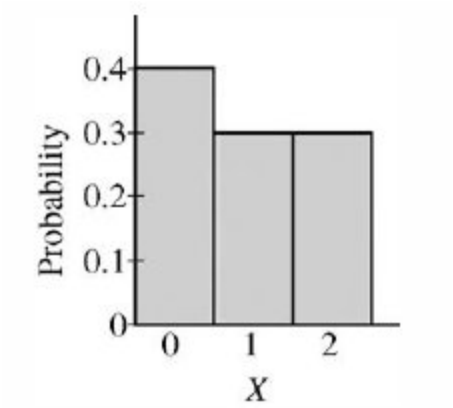
(D) $\frac{396}{538}$ ✓

Answer D

Correct. There were $956 - 418 = 538$ seniors who gave a response other than college. Of those, $99 + 49 + 63 = 211$ were from High School A and $83 + 31 + 71 = 185$ were from High School C. Thus $\frac{211+185}{538} = \frac{396}{538}$ is the probability that a senior selected will not be from High School B given that the senior gave a response other than college.

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13. The number of points, X, scored in a game has the probability distribution below



The number of points obtained in one game is independent of the number of points obtained in a second game. When the game is played twice, the sum of the number of points for both times could be 0, 1, 2, 3, or 4. If Y represents the sampling distribution of the sum of the scores when the game is played twice, for which value of Y will the probability be greatest?

(A) 0

(B) 2 ✓

(C) 3

(D) 4

14. The SC Electric Company has bid on two electrical wiring jobs. The owner of the company believes that

- the probability of being awarded the first job (event A) is 0.75;
- the probability of being awarded the second job (event B) is 0.5; and
- the probability of being awarded both jobs (event (A and B)) is 0.375.

If the owner's beliefs are correct, which of the following statements must be true concerning event A and event B ?

(A) Event A and event B are mutually exclusive and are independent.

(B) Event A and event B are mutually exclusive and are <u>not</u> independent.

(C) Event A and event B are <u>not</u> mutually exclusive and are independent. ✓

(D) Event A and event B are <u>not</u> mutually exclusive and are <u>not</u> independent.

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15. For which of the following probability assignments are events A and B independent?

(A) $P(A \cap B^c) = 0.3$, $P(A \cap B) = 0.12$, and $P(A^c \cap B) = 0.4$.

(B) $P(A \cap B^c) = 0.3$, $P(A \cap B) = 0.3$, and $P(A^c \cap B) = 0.3$.

(C) $P(A \cap B^c) = 0.1$, $P(A \cap B) = 0.1$, and $P(A^c \cap B) = 0.4$. ✓

(D) $P(A \cap B^c) = 0.3$, $P(A \cap B) = 0.0$, and $P(A^c \cap B) = 0.2$.

Answer C

Correct. Events A and B are independent if $P(A \cap B) = [P(A)][P(B)]$. Here, $P(A) = P(A \cap B) + P(A \cap B^c) = 0.1 + 0.1 = 0.2$ and $P(B) = P(B \cap A) + P(B \cap A^c) = 0.1 + 0.4 = 0.5$. Therefore, $[P(A)][P(B)] = (0.2)(0.5) = 0.1$ and $0.1 = P(A \cap B)$.

16. At a small company, 40 percent of the employees are classified as junior level, and 32 percent of the employees work in department A. Of those classified as junior level, 55 percent work in department A. One employee will be selected at random.

What is the probability that the selected employee works in department A and is not classified as junior level?

(A) 0.10 ✓

(B) 0.19

(C) 0.22

(D) 0.32

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17. A company is considering purchasing the mineral rights to two different mountains. The probability that it will purchase the mineral rights to the first mountain is 0.55. The probability that it will purchase the mineral rights to the second mountain is 0.4. Assuming the decisions to purchase the mineral rights to each mountain are made independently, what is the probability that it will purchase the mineral rights to exactly one of the two mountains?

(A) 0.22

(B) 0.33

(C) 0.51 ✓

(D) 0.95

Answer C

Correct. If F represents the event that the mineral rights to the first mountain are purchased and S represents the event that the mineral rights to the second mountain are purchased, then
$P(F \cap S^c) + P(F^c \cap S) = [P(F)][P(S^c)] + [P(F^c)][P(S)] = (0.55)(0.6) + (0.45)(0.4) = 0.51.$

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18. Students at a local elementary school were shown a painting and asked which emotion—joy, happiness, love, or anger—they felt by looking at the painting. The students were classified by their age. The following table summarizes the responses of the students by age-group.

	Joy	Happiness	Love	Anger	Total
6 to 8 years	28	20	40	18	106
9 to 11 years	61	25	80	60	226
Total	89	45	120	78	332

One student from the school will be selected at random. What is the probability that the student is in the age-group of 6 to 8 years given that the selected student responded joy?

(A) $\frac{28}{332}$

(B) $\frac{28}{106}$

(C) $\frac{106}{332}$

(D) $\frac{28}{89}$ ✓

Answer D

Correct. Of the 89 students who responded joy, 28 were in the age-group of 6 to 8 years, so $\frac{28}{89}$ is the probability that a student will be in the age-group of 6 to 8 years given that the student responded joy.

19. Ninety percent of the people who have a particular disease will have a positive result on a given diagnostic test. Ninety percent of the people who do not have the disease will have a negative result on this test. If 5 percent of a certain population has the disease, what percent of that population would test positive for the disease?

(A) 4.5%

(B) 5%

(C) 14% ✓

(D) 90%

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20. Mating eagle pairs typically have two baby eagles (called eaglets). When there are two eaglets, the parents always feed the older eaglet until it has had its fill, and then they feed the younger eaglet. This results in an unequal chance of survival for the two eaglets. Suppose that the older eaglet has a 50 percent chance of survival. If the older eaglet survives, the younger eaglet has a 10 percent chance of survival. If the older eaglet does not survive, the younger eaglet has a 30 percent chance of survival. Let X be the number of eaglets that survive. Which of the following tables shows the probability distribution of X?

(A)

x	0	1	2
$p(x)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

(B) ✓

x	0	1	2
$p(x)$	0.35	0.60	0.05

(C)

x	0	1	2
$p(x)$	0.05	0.90	0.05

(D)

x	0	1	2
$p(x)$	0.10	0.30	0.50

21. According to a report for veterinarians in the United States, 36.5 percent of households in the United States own dogs and 30.4 percent of households in the United States own cats. If one household in the United States is selected at random, what is the probability that the selected household will own a dog or a cat?

(A) 0.331

(B) 0.558

(C) 0.669

(D) Not enough information is given to determine the probability. ✓